

24-July-SSM manager

AWS Systems Manager (SSM) is a service that helps you manage EC2 instances and on-premises servers without logging in using SSH (Linux) or RDP (Windows). SSM is a secure remote management service for servers.

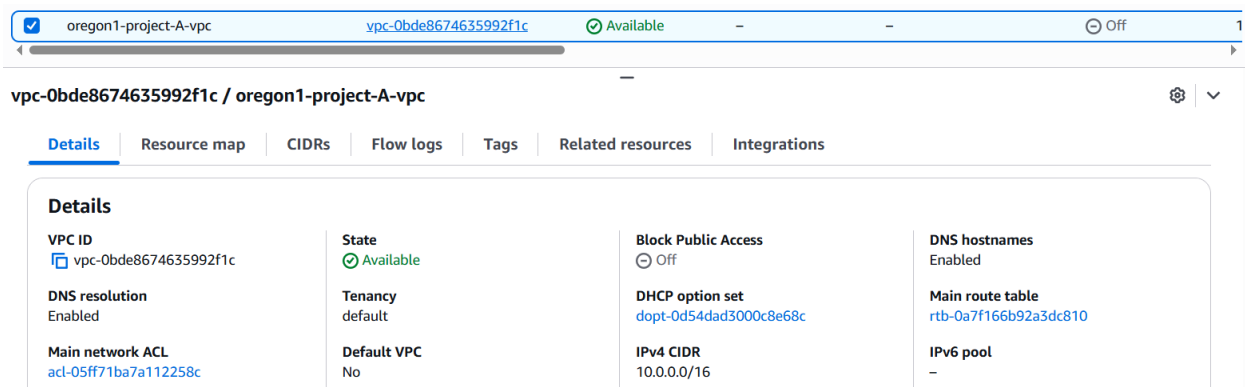
- Ensure SSM agent should run, we need to attach SSM manager permission to “AmazonSSMManagedInstanceCore” role.
- Required -> HTTPs protocol, SSM agent, Outbound .
- Not required -> port 22, Bastion Host, keys

Advantages:

- No need to open port 22 (SSH) or 3389 (RDP).
- Secure access using IAM-based access control
- Full session logging
- No SSH keys required.
- No Bastion Host required.
- Works on private EC2 instances
- Supports automation

LAB: Accessing EC2 Using SSM Session Manager and Local CLI

- Create VPC & more -> 1az – 1 private and 1 public – NAT=required
vpc endpoint-none, CIDR = 10.0.0.0/16 – united state(Oregon)



- create a role with “AmazonSSMManagedInstanceCore” role.



24-July-SSM manager

Step 1: Select trusted entities

[Edit](#)

Trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "ec2.amazonaws.com"
12        ]
13      }
14    ]
15  }
16 }
```

Step 2: Add permissions

[Edit](#)

Permissions policy summary

Policy name ↗	Type	Attached as
AmazonSSMManagedInstanceCore	AWS managed	Permissions policy

- Launch private ec2 instance.

Name and tags [Info](#)

Name

myprivt

[Add additional tags](#)

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-077d4c57ecdca57e6 (64-bit (x86), uefi-preferred) / ami-07e4bc5be76f861b0 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.12.20260724.0 x86_64 HVM kernel-6.18

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-077d4c57ecdca57e6

Creation Date

2026-07-24

Username

ec2-user

Verified provider

▼ Instance type

[Info](#) | [Get advice](#)

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour On-Demand Linux base pricing: 0.0104 USD per Hour

All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login)

[Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select

[Create new key pair](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-077d4c57ecdca57e6

▼ Summary

Number of instances

[Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-077d4c57ecdca57e6

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

[Preview code](#)

- PVT – amzLinux, autoAssignIP=disable{key not required}, Advanced details = IAM instance details = select your role

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24-July-SSM manager

▼ Network settings

Info

VPC - required

Info

vpc-0bde8674635992f1c (oregon1-project-A-vpc)

10.0.0.0/16

Subnet

Info

subnet-03eade8dc9073a70

oregon1-project-A-subnet-private1-us-west-2a

VPC: vpc-0bde8674635992f1c

Owner: 204807767462

Availability Zone: us-west-2a (usw2-az2)

Zone type: Availability Zone

IP addresses available: 4091

CIDR: 10.0.128.0/20

Auto-assign public IP

Info

Disable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

ssm-sg

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./!@#%&'()*+,-=:;[]^_`{|}~

Description - required

Info

ssm-sg-agent

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type

Info

ssh

Protocol

Info

TCP

Port range

Info

22

Source type

Info

Anywhere

Source

Info

Q Add CIDR, prefix list or security group

Description - optional

Info

e.g. SSH for admin desktop

▼ Advanced details

Info

Domain join directory

Info

Select

IAM instance profile

Info

ssm-role

arn:aws:iam::204807767462:instance-profile/ssm-role

Hostname type

Info

IP name

DNS Hostname

Info

☒ Enable IP name IPv4 (A record) DNS requests

☐ Enable resource-based IPv4 (A record) DNS requests

☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery

Info

Select

Shutdown behavior

Info

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12...read more

ami-077d4c57ecdca57e6

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12...read more

ami-077d4c57ecdca57e6

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

- Lets connect Ec2 instance -> SSM agen status = should be Online(first verify)
SSM session manager-TAB -> Connect

[EC2](#) > [Instances](#) > [i-09e562bb1e2f7ae1d](#) > [Connect to instance](#)

Connect Info

Connect to an instance using the browser-based client.

Instance ID i-09e562bb1e2f7ae1d (myprivt)	VPC ID vpc-0bde8674635992f1c	Security groups sg-0ec85af681f410b583 (ssm-sg)	IAM role ssm-role
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EC2 Instance Connect

SSM Session Manager

SSH client

EC2 serial console

SSM agent status

Ping status Online	Last ping time Sat Jul 25 2026 10:52:59 GMT+0530 (India Standard Time)	Session Manager connection status Connected	Agent version 3.3.4624.0
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Session Manager usage:

- Connect to your instance without SSH keys, a bastion host, or opening any inbound ports.
- Sessions are secured using an AWS Key Management Service key.
- You can log session commands and details in an Amazon S3 bucket or CloudWatch Logs log group.
- Configure sessions on the Session Manager [Preferences](#) page.

Cancel

Connect

- After Login we can verify SSM agent from Terminal
systemctl status amazon-ssm-agent (not required)

24-July-SSM manager

```
[root@ip-10-0-133-98 ~]# systemctl status amazon-ssm-agent
● amazon-ssm-agent.service - amazon-ssm-agent
   Loaded: loaded (/usr/lib/systemd/system/amazon-ssm-agent.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-07-25 05:22:55 UTC; 5min ago
   Main PID: 1885 (amazon-ssm-agent)
     Tasks: 30 (limit: 1059)
   Memory: 91.6M
      CPU: 1.165s
   CGroup: /system.slice/amazon-ssm-agent.service
           └─1885 /usr/bin/amazon-ssm-agent
             └─1960 /usr/bin/ssm-agent-worker
               └─2104 /usr/bin/ssm-session-worker root-xu569x4df6ekbkenuix6e9tp4i
                 └─2126 sh

Jul 25 05:22:57 ip-10-0-133-98.us-west-2.compute.internal amazon-ssm-agent[1885]: 2026-07-25 05:22:57.4497 INFO [amazon-ssm-agent] [LongRunningWorkerContainer] [WorkerProvider] Worker ssm-
Jul 25 05:22:57 ip-10-0-133-98.us-west-2.compute.internal amazon-ssm-agent[1885]: 2026-07-25 05:22:57.4497 INFO [amazon-ssm-agent] [LongRunningWorkerContainer] Monitor long running worker s
Jul 25 05:23:58 ip-10-0-133-98.us-west-2.compute.internal amazon-ssm-agent[1885]: 2026-07-25 05:23:58.2587 INFO [CredentialRefresher] Next credential rotation will be in 28.969585052183334
Jul 25 05:24:58 ip-10-0-133-98.us-west-2.compute.internal amazon-ssm-agent[1885]: 2026-07-25 05:24:58.2565 INFO [CredentialRefresher] Next credential rotation will be in 27.969620493616667
Jul 25 05:25:58 ip-10-0-133-98.us-west-2.compute.internal amazon-ssm-agent[1885]: 2026-07-25 05:25:58.2821 INFO [CredentialRefresher] Next credential rotation will be in 26.969193723266667
Jul 25 05:26:24 ip-10-0-133-98.us-west-2.compute.internal useradd[2116]: new group: name=ssm-user, GID=1001
Jul 25 05:26:58 ip-10-0-133-98.us-west-2.compute.internal amazon-ssm-agent[1885]: 2026-07-25 05:26:58.2666 INFO [CredentialRefresher] Next credential rotation will be in 25.969452320983333
Jul 25 05:27:47 ip-10-0-133-98.us-west-2.compute.internal sudo[2185]: ssm-user : TTY=pts/0 ; FWD=/usr/bin ; USER=root ; COMMAND=/usr/bin/su -
Jul 25 05:27:58 ip-10-0-133-98.us-west-2.compute.internal amazon-ssm-agent[1885]: 2026-07-25 05:27:58.2574 INFO [CredentialRefresher] Next credential rotation will be in 24.969606205133333
lines 1-23/23 (END)
```

- Remove inbound rule and check, are we able to connect or not – yes we can able to connect only required outbound rule to connect.

sg-0ec85a681f410b583 - ssm-sg

Actions ▾

Details

Security group name ssm-sg	Security group ID sg-0ec85a681f410b583	Description ssm-sg-agent	VPC ID vpc-0bde8674635992f1c
Owner 204807767462	Inbound rules count 0 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
No security group rules found						

Session ID: root-xu569x4df6ekbkenuix6e9tp4i | Shortcuts | Instance ID: i-09e562bb1e2f7ae1d (mypvt) | Terminate session

```
sh-5.2$
sh-5.2$
sh-5.2$
sh-5.2$ sudo su -
[root@ip-10-0-133-98 ~]#
```

So far we have connected via NAT for ssm-session.

We can also achieve the ssm-session with VPC-endpoint thus Aws system manager is lies outside of vpc service but in aws.

- We required to create 3 endpoints we need to attach and enable
 - 1) Connect the SSM agent to the system manager - **.ssm**
 - 2) Handles standard agent to service communication and management commands - **.ec2messages**
 - 3) Required explicitly for Establishing secure interactive data channels with session manager - **.ssmmessages**

Lets create Endpoints.

Path is vpc -> endpoints - >create

- Name = ssm-endpoint , Aws services

Create endpoint Info

Create the type of VPC endpoint that supports the service, service network or resource to which you want to connect.

Endpoint settings

Specify a name and select the type of endpoint.

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

ssm-endpoint

Type Info
Select a category

☒ **AWS services**
Connect to services provided by Amazon with an interface endpoint, or a Gateway endpoint

☐ **PrivateLink Ready partner services**
Connect to SaaS services which have AWS Service Ready designation with an Interface endpoint. Uses AWS PrivateLink

☐ **AWS Marketplace services**
Connect to SaaS services that you have purchased through AWS Marketplace with an Interface Endpoint

☐ **EC2 Instance Connect Endpoint**
An elastic network interface that allows you to connect to resources in a private subnet

☐ **Resources**
Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS PrivateLink

☐ **Service networks**
Connect to VPC Lattice service networks with a Service network endpoint. Uses AWS PrivateLink

☐ **Endpoint services that use NLBs and GWLBs**
Find services shared with you by service name. Connect to a Network LoadBalancer (NLB) service with an interface endpoint or to a Gateway LoadBalancer (GWLB) service with a Gateway Load Balancer endpoint

Service Region

☐ **Enable Cross Region endpoint** Info
Connect to cross Region enabled services.

service = SSM - interface

vpc = select yours

subnets = server running(private)

Services (1/1)

Search

Service Name = com.amazonaws.us-west-2:ssm X Clear filters

Service Name	Owner	Type
com.amazonaws.us-west-2:ssm	amazon	Interface

Network settings

Select the VPC in which to create the endpoint

VPC
Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

vpc-0bde8674635992f1c (oregon1-project-A-vpc)

Additional settings

Private DNS name

☒ **Enable private DNS name** Info
Associates a private hosted zone with the VPC that contains a record set that enables you to leverage Amazon's private network connectivity to the service while making requests to the service's default public endpoint DNS name. To use this feature, ensure that the attributes 'Enable DNS hostnames' and 'Enable DNS support' are enabled for your VPC.

DNS record IP type

☒ IPv4
☐ IPv6
☐ Dualstack
☐ Service defined

Subnets (1/4) Info

Availability Zone	Subnet ID	Designate IP addresses	IPv4 address	IPv6 address
☒ usw2-az2 (us-west-2a)	subnet-03eade8dcc9073a70	☐		

24-July-SSM manager

select security group = ssm-sg

IP address type
☒ IPv4
☐ IPv6
☐ Dualstack

Security groups (1/2) [Info](#)

Q Search

VPC ID: vpc-0bde8674635992f1c [X](#) [Clear filters](#)

< 1 > [⚙](#)

☒	Group ID	Group name	VPC ID	Description
☐	sg-0f9b7aa8d4fcc4a4f	default	vpc-0bde8674635992f1c	default VPC security group
☒	sg-0ec85a681f410b583	ssm-sg	vpc-0bde8674635992f1c	ssm-sg-agent

sg-0ec85a681f410b583 [X](#)

- Name = messagessm-endpoint , Aws services

Endpoint settings
Specify a name and select the type of endpoint.

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

messagessm-endpoint

Type [Info](#)
Select a category

☒ **AWS services**
Connect to services provided by Amazon with an interface endpoint, or a Gateway endpoint

☐ **PrivateLink Ready partner services**
Connect to SaaS services which have AWS Service Ready designation with an interface endpoint. Uses AWS PrivateLink

☐ **AWS Marketplace services**
Connect to SaaS services that you have purchased through AWS Marketplace with an interface endpoint

service = .SSMmessages - interface

vpc = select yours

subnets = server running(private)

Services (1/1) [Info](#)

Q Search

Service Name: com.amazonaws.us-west-2.ssmmessages [X](#) [Clear filters](#)

< 1 > [⚙](#)

☒	Service Name	Owner	Type
☒	com.amazonaws.us-west-2.ssmmessages	amazon	Interface

Network settings
Select the VPC in which to create the endpoint.

VPC
Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

vpc-0bde8674635992f1c (oregon1-project-A-vpc) [⌵](#) [⌵](#)

Additional settings

Private DNS name
☒ **Enable private DNS name** [Info](#)
Associates a private hosted zone with the VPC that contains a record set that enables you to leverage Amazon's private network connectivity to the service while making requests to the service's default public endpoint DNS name. To use this feature, ensure that the attributes 'Enable DNS hostnames' and 'Enable DNS support' are enabled for your VPC.

DNS record IP type
☒ IPv4
☐ IPv6
☐ Dualstack
☐ Service defined

Subnets (1/4) [Info](#)

☒ Availability Zone ☒ Subnet ID ☐ Designate IP addresses ☐ IPv4 address ☐ IPv6 address

☒ usw2-az2 (us-west-2a) [⌵](#) subnet-03eade8dcc9073a70 ☐ ☐

select security group = ssm-sg

Security groups (1/2) [Info](#)

Q Search

VPC ID: vpc-0bde8674635992f1c [X](#) [Clear filters](#)

< 1 > [⚙](#)

☒	Group ID	Group name	VPC ID	Description
☐	sg-0f9b7aa8d4fcc4a4f	default	vpc-0bde8674635992f1c	default VPC security group
☒	sg-0ec85a681f410b583	ssm-sg	vpc-0bde8674635992f1c	ssm-sg-agent

sg-0ec85a681f410b583 [X](#)

24-July-SSM manager

- Name = ec2messages-endpoint , Aws services

Endpoint settings
Specify a name and select the type of endpoint.

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

ec2messages-endpoint

Type [Info](#)
Select a category

☒ **AWS services**
Connect to services provided by Amazon with an interface endpoint, or a Gateway endpoint

☐ **PrivateLink Ready partner services**
Connect to SaaS services which have AWS Service Ready designation with an Interface endpoint. Uses AWS PrivateLink

☐ **AWS Marketplace services**
Connect to SaaS services that you have purchased through AWS Marketplace with an Interface Endpoint

service = SSM ec2messages - interface

vpc = select yours

subnets = server running(private)

Services (1/1) [Info](#)

Q Search

Service Name = com.amazonaws.us-west-2.ec2messages X Clear filters

Service Name	Owner	Type
com.amazonaws.us-west-2.ec2messages	amazon	Interface

Network settings
Select the VPC in which to create the endpoint

VPC
Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

vpc-0bde8674635992f1c (oregon1-project-A-vpc)

Additional settings

Private DNS name
☒ Enable private DNS name [Info](#)
Associates a private hosted zone with the VPC that contains a record set that enables you to leverage Amazon's private network connectivity to the service while making requests to the service's default public endpoint DNS name. To use this feature, ensure that the attributes 'Enable DNS hostnames' and 'Enable DNS support' are enabled for your VPC.

DNS record IP type
☒ IPv4
☐ IPv6
☐ Dualstack
☐ Service defined

Subnets (1/4) [Info](#)

Availability Zone	Subnet ID	Designate IP addresses	IPv4 address	IPv6 address
☒ usw2-az2 (us-west-2a)	subnet-03eade8dc9073a70	☐		
☐ usw2-az1 (us-west-2b)	No subnet available	☐		
☐ usw2-az3 (us-west-2c)	No subnet available	☐		
☐ usw2-az4 (us-west-2d)	No subnet available	☐		

select security group = ssm-sg

Security groups (1/2) [Info](#)

Q Search

VPC ID : vpc-0bde8674635992f1c X Clear filters

Group ID	Group name	VPC ID	Description
☐ sg-0f9b7aa8d4fcc4a4f	default	vpc-0bde8674635992f1c	default VPC security group
☒ sg-0ec85a681f410b583	ssm-sg	vpc-0bde8674635992f1c	ssm-sg-agent

sg-0ec85a681f410b583 X

- This will create 3 endpoints

Endpoints (3) [Info](#)

Q Find endpoints by attribute or tag

Name	VPC endpoint ID	Endpoint type	Status	Service name	Service network	Resource configuration
☐ ssm-endpoint	yocce-0409ef2ec5a4497c1	Interface	Available	com.amazonaws.us-west-2.ssm	-	-
☐ messagessm-endpoint	yocce-056b2904c8d5c8801	Interface	Available	com.amazonaws.us-west-2.ssmmessages	-	-
☐ ec2messages-endpoint	yocce-038420823cc39a08a	Interface	Available	com.amazonaws.us-west-2.ec2messages	-	-

24-July-SSM manager

Try to connect Again the private ec2 instance and remove NAT gateway so we can ensure that connection is done only from endpoints.

Close on-going session(older) if running

Again connect Via SSM-Manager verify the status of agent online status.

[EC2](#) > [Instances](#) > [i-09e562bb1e2f7ae1d](#) > [Connect to instance](#)

Connect info

Connect to an instance using the browser-based client.

Instance ID i-09e562bb1e2f7ae1d (mypvt)	VPC ID vpc-0bde8674635992f1c	Security groups sg-0ec85a681f410b583 (ssm-sg)	IAM role ssm-role
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[EC2 Instance Connect](#) | [SSM Session Manager](#) | [SSH client](#) | [EC2 serial console](#)

SSM agent status

Ping status Online	Last ping time Sat Jul 25 2026 12:16:08 GMT+0530 (India Standard Time)	Session Manager connection status Connected	Agent version 3.3.4624.0
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sg-0ec85a681f410b583

sg-0ec85a681f410b583 - ssm-sg

[Actions](#)

Details

Security group name ssm-sg	Security group ID sg-0ec85a681f410b583	Description ssm-sg-agent	VPC ID vpc-0bde8674635992f1c
Owner 204807767462	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

[Inbound rules](#) | [Outbound rules](#) | [Sharing](#) | [VPC associations](#) | [Related resources - new](#) | [Tags](#)

Inbound rules (1)

[Manage tags](#) [Edit inbound rules](#)

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-0e0e1f2b851d15732	IPv4	HTTPS	TCP	443	0.0.0.0/0

```
sh-5.2$ sudo su -
Last login: Sat Jul 25 05:27:47 UTC 2026 on pts/1
[root@ip-10-0-133-98 ~]#
```

remove NAT gateway so we can ensure that connection is done only from endpoints.

rtb-003f3f3da01ceba1c

Updated routes for rtb-003f3f3da01ceba1c / oregon1-project-A-rtb-private1-us-west-2a successfully

rtb-003f3f3da01ceba1c / oregon1-project-A-rtb-private1-us-west-2a

[Actions](#)

Details info

Route table ID rtb-003f3f3da01ceba1c	Main No	Explicit subnet associations subnet-03eade8dc9073a70 / oregon1-project-A-subnet-private1-us-west-2a	Edge associations -
VPC vpc-0bde8674635992f1c oregon1-project-A-vpc	Owner ID 204807767462		

[Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#) | [Related resources](#)

Routes (1)

[Filter routes](#)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

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```
sh-5.2$ sudo su -  
Last login: Sat Jul 25 05:27:47 UTC 2026 on pts/1  
[root@ip-10-0-133-98 ~]#  
[root@ip-10-0-133-98 ~]#  
[root@ip-10-0-133-98 ~]#  
[root@ip-10-0-133-98 ~]#  
[root@ip-10-0-133-98 ~]#  
[root@ip-10-0-133-98 ~]#  
[root@ip-10-0-133-98 ~]#  
[root@ip-10-0-133-98 ~]# without NAT
```